Development of Vertical Takeoff and Landing Fixed-Wing Tilt-Rotor Craft and Trajectory for Wildfire Fighting

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I. Nomenclature

Frames, operators, and indices

$\mathcal{N}$	=	inertial/navigation frame
$\mathcal{B}$	=	body-fixed frame attached to the vehicle center of mass
$SO(3)$	=	special orthogonal group of valid rotation matrices
R_{bn}	=	direction cosine matrix mapping body-frame vectors to inertial-frame vectors
I	=	identity matrix
$\mathrm{vec}(\cdot)$	=	matrix vectorization operator
$(\cdot)^\top$	=	transpose
$\det(\cdot)$	=	determinant
$\mathrm{tr}(\cdot)$	=	trace
$\Vert \cdot \Vert$	=	Euclidean norm
$\Vert \cdot \Vert_F$	=	Frobenius norm
$\widehat{(\cdot)}$	=	skew-symmetric matrix operator
i	=	rotor index or NMPC model index, depending on context
j	=	control-surface or HOCBF tube-channel index
N_u	=	number of control inputs

Vehicle states and kinematics

x	=	full-order VTOL state vector
x_r	=	24-state reduced-order NMPC prediction state
x_α	=	19-state attitude–velocity–rate NMPC prediction state
q_g	=	generalized coordinate vector
ν	=	generalized speed vector
$r_N = [x_N, y_N, z_N]^\top$	=	inertial position vector
v_N	=	inertial translational velocity vector
$v_B = [u_b, v_b, w_b]^\top$	=	body-frame translational velocity vector
$\omega_B = [p, q, r]^\top$	=	body angular velocity vector
p, q, r	=	roll, pitch, and yaw body rates
$s = [s_x, s_y, s_z]^\top$	=	internal slosh displacement in body coordinates
$\dot{s}$	=	internal slosh velocity
β_i	=	rotor tilt angle for rotor i
$\beta = [\beta_1, \beta_2]^\top$	=	rotor tilt-angle vector

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$\dot{\beta}_i$	=	rotor tilt rate for rotor i
T_i	=	thrust state of rotor i
$T = [T_1, T_2]^\top$	=	rotor thrust-state vector
u	=	NMPC control input vector
u_k	=	NMPC control input at prediction node k
$u_k^\star$	=	optimized NMPC control input at node k
$T_{\text{cmd},i}$	=	commanded thrust for rotor i
$\beta_{\text{cmd},i}$	=	commanded tilt angle for rotor i
δ_e	=	elevator deflection command
δ_a	=	aileron deflection command
δ_r	=	rudder deflection command
$\delta = [\delta_e, \delta_a, \delta_r]^\top$	=	aerodynamic control-surface command vector

Vehicle, rotor, and actuator parameters

m	=	total aircraft mass
m_s	=	slosh modal mass
S	=	wing reference area
b	=	wing span
$\bar{c}$	=	mean aerodynamic chord
x_a, y_a, z_a	=	rotor position offsets from the vehicle center of mass
r_i	=	position vector from the center of mass to rotor i
$t_i(\beta_i)$	=	thrust-direction unit vector for rotor i
$T_{\max}$	=	maximum rotor thrust
τ_T	=	thrust actuator time constant
τ_β	=	rotor-tilt actuator time constant
$F_{T_i,B}$	=	body-frame thrust force from rotor i
$F_{T,B}$	=	total body-frame rotor thrust force
$\tau_{T,B}$	=	thrust-induced moment about the center of mass
Q_i	=	reaction torque generated by rotor i
c_{Q_i}	=	rotor torque-per-thrust coefficient
σ_i	=	rotor spin-direction sign
$\tau_{Q,B}$	=	total rotor reaction torque

Aerodynamics and local flow

V	=	airspeed magnitude
α	=	angle of attack
β_a	=	aerodynamic sideslip angle
ρ	=	air density
$\bar{q}$	=	freestream dynamic pressure
$\widehat{p}, \widehat{q}, \widehat{r}$	=	nondimensional roll, pitch, and yaw rates
L, D, Y	=	lift, drag, and side force
F_w	=	aerodynamic force vector in wind axes
$F_{a,B}$	=	aerodynamic force vector in body axes
$\tau_{a,B}$	=	aerodynamic moment vector in body axes
C_L, C_D, C_Y	=	lift, drag, and side-force coefficients
C_ℓ, C_m, C_n	=	rolling, pitching, and yawing moment coefficients
$C_{\bullet 0}$	=	zero-angle or baseline aerodynamic coefficient, where $\bullet \in \{L, D, m\}$
$C_{\bullet\alpha}$	=	angle-of-attack derivative of coefficient $\bullet$
$C_{\bullet\beta}$	=	sideslip derivative of coefficient $\bullet$
$C_{\bullet p}, C_{\bullet q}, C_{\bullet r}$	=	body-rate derivatives of aerodynamic coefficient $\bullet$
$C_{\bullet\delta_e}, C_{\bullet\delta_a}, C_{\bullet\delta_r}$	=	control-surface derivatives of aerodynamic coefficient $\bullet$
k_D	=	parabolic drag-polar factor
$R_x(\cdot), R_y(\cdot), R_z(\cdot)$	=	elementary rotation matrices about the x-, y-, and z-axes, respectively
r_j	=	position vector from the center of mass to control surface j
$v_{B,j}$	=	local body-frame velocity at control surface j

$v_{\text{prop},j}$ = rotor-induced slipstream velocity at control surface j
$V_{\text{eff},j}$ = local effective airspeed at control surface j
$\bar{q}_j$ = local dynamic pressure at control surface j
A_i = rotor disk area for rotor i
η_{ij} = slipstream impingement factor from rotor i to surface j
$\Delta F_{a,B}$ = control-surface-induced aerodynamic force increment
$\Delta \tau_{a,B}$ = control-surface-induced aerodynamic moment increment

Slosh and full-order dynamics

$F_{s,B}$ = restoring and damping force acting on the slosh mass
k_x, k_y, k_z = slosh stiffness coefficients
c_x, c_y, c_z = slosh damping coefficients
$M(q_g)$ = generalized mass matrix from Kane's equations
$F(q_g, \nu, \cdot)$ = generalized forcing vector
$\Gamma(q_g, \nu)$ = kinematic mapping from generalized speeds to generalized-coordinate rates
$f(x)$ = nonlinear drift vector field
$G(x)$ = nonlinear input matrix
$G_T(x)$ = thrust-command input mapping
$G_\beta(x)$ = tilt-command input mapping
$G_\delta(x)$ = aerodynamic control-surface input mapping
$d(x,t)$ = lumped disturbance, model mismatch, and unmodelled dynamics
$F_0(x)$ = generalized force independent of current control-surface deflections
$F_{\text{Kane},0}$ = Kane forcing vector evaluated with zero control-surface deflection
$B_\delta(x)$ = generalized control-surface effectiveness matrix
$B_{F,\delta}(x)$ = aerodynamic force-effectiveness block
$B_{\tau,\delta}(x)$ = aerodynamic moment-effectiveness block
$J_W(q_g)$ = Kane generalized-force mapping from body-frame wrench to generalized forces
$\Delta_\delta(x, \delta)$ = non-affine aerodynamic residual associated with control-surface deflections
$f_d(x_k, u_k)$ = discrete-time prediction dynamics used by the NMPC

Reference trajectory generation

t = time
t_0, t_f = initial and final times of a scheduled flight phase
T_p = phase duration, $T_p = t_f - t_0$
$\xi_r(t)$ = scheduled reference variable used for minimum-jerk phase generation
ξ_0, ξ_f = initial and final values of the scheduled reference variable for one phase
J_ξ = squared-jerk cost associated with the scheduled reference variable $\xi_r(t)$
$\dot{\xi}_r, \ddot{\xi}_r, \dddot{\xi}_r$ = first, second, and third time derivatives of the scheduled reference variable
$\xi_r^{(6)}$ = sixth time derivative of the scheduled reference variable appearing in the Euler–Lagrange minimum-jerk condition
λ = normalized phase coordinate, $\lambda = (t - t_0)/T_p$, with $\lambda \in [0, 1]$
$\sigma(\lambda)$ = quintic minimum-jerk blending function, $\sigma(\lambda) = 10\lambda^3 - 15\lambda^4 + 6\lambda^5$
$\sigma'(\lambda), \sigma''(\lambda), \sigma'''(\lambda)$ = first, second, and third derivatives of the minimum-jerk blending function with respect to λ
$r_{N,r} = [x_{N,r}, y_{N,r}, z_{N,r}]^\top$ = inertial reference position vector
V_r = reference airspeed used by the trajectory generator
ψ_r = reference heading angle
$\dot{\psi}_r$ = reference heading rate
ϕ_r = reference roll angle or coordinated-turn bank angle
θ_r = reference pitch angle

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θ_r = reference pitch angle
$\phi_{\max}$ = prescribed bank-angle limit used when clipping the coordinated-turn roll command
γ_r = reference flight-path angle
α_{trim} = trimmed angle of attack used when computing the pitch reference
$R_{bn,r}$ = reference direction cosine matrix mapping body-frame vectors to inertial-frame vectors
$x_{N,\mathrm{hov}}, y_{N,\mathrm{hov}}, z_{N,\mathrm{hov}}$ = inertial hover position used before the final landing phase
$z_{N,\mathrm{td}}$ = inertial touchdown vertical position
$\dot{z}_{N,r}, \ddot{z}_{N,r}$ = reference vertical velocity and acceleration during the landing phase
g = gravitational acceleration

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II. Introduction

Tilt-rotor and transitional VTOL aircraft combine rotor-borne vertical flight with wing-borne cruise. This architecture offers runway-independent takeoff and landing while retaining efficient fixed-wing flight at speed. The difficulty is that the vehicle must pass through a transition corridor where the dominant sources of lift, propulsive force, and control authority change continuously. Classical tilt-rotor flight-test programs such as the XV-15 demonstrated the feasibility of this conversion concept, but they also made clear that transition flight is governed by strong coupling among rotor thrust, nacelle or rotor tilt, wing aerodynamics, rotor–wing interaction, actuator limits, and vehicle attitude dynamics [1]. Modern small and unmanned tilt-rotor systems inherit the same modelling challenge: the controller must coordinate rotor tilt, thrust, attitude, and aerodynamic surfaces without introducing abrupt mode-switching transients [2, 3].

The central challenge in modelling this flight regime is the coexistence of rotorcraft and fixed-wing effects. In hover and low-speed flight, the vehicle is primarily supported by rotor thrust, and the flow field is strongly influenced by rotor-induced velocity, slipstream impingement, and ground effect. In cruise, the wing and conventional aerodynamic surfaces dominate lift and moment generation, and the rotors become primarily propulsive. The conversion phase between these regimes contains oblique rotor inflow, changing slipstream coherence, stall or post-stall wing conditions, and a rapidly changing allocation of vertical and forward force. High-fidelity methods based on blade-element aerodynamics, free-wake modelling, or computational fluid dynamics can resolve more of these effects, but they are often too expensive and parameter-intensive for real-time control design and trajectory generation [4, 5]. Conversely, a purely kinematic or point-mass model is insufficient because it cannot represent the actuator and force-vector limitations that determine whether a transition trajectory is trackable.

This paper, therefore, adopts a reduced-order, control-oriented modelling strategy. The airframe is represented as a rigid body with a moving slosh mass, the two rotors are represented as tilting thrust application points offset from the center of mass, and the aerodynamic model uses quasi-steady fixed-wing stability derivatives modified by transition-specific effects. This modelling level follows the same philosophy as recent tilt-rotor multibody simulation work: retain the dominant vehicle-level force and moment couplings in a compact equation set, while avoiding blade-resolved wake states unless they are required by the control objective [6]. Kane's method is used because it naturally accommodates generalized speeds, moving particles, offset loads, and nonconservative forces in mass-matrix form [7, 8]. The symbolic derivation is then evaluated numerically in the plant right-hand side.

The second objective is to construct a reference trajectory suitable for nonlinear model predictive control (NMPC). Transition references must be smooth enough to respect finite thrust lag, finite rotor-tilt bandwidth, and aerodynamic-surface rate limits. A discontinuous or overly aggressive schedule can demand an instantaneous change in net force vector, leading to tilt saturation, thrust saturation, attitude-rate spikes, or large tracking error. Minimum-jerk blending is used as a closed-form smoothing primitive because it provides zero endpoint rate and acceleration for scheduled variables such as rotor tilt, altitude, heading, airspeed, and hover/cruise blending [9]. This choice is consistent with polynomial trajectory-generation literature for aerial robots, where high-order smoothness is used to make references more compatible with finite actuator bandwidth and dynamic constraints [10–12].

Minimum jerk is not treated as a guarantee of complete feasibility. A smooth schedule can still be dynamically impossible if it requires more thrust, lift, drag compensation, or elevator trim than the vehicle can supply. Therefore, the reference generator combines minimum-jerk phase scheduling with a lower-order aerodynamic and propulsive force-balance calculation. At each time step, the generator estimates lift and drag, imposes a feasible rotor-tilt floor for vertical support, computes the thrust required for vertical and forward force balance, clips thrust to physical limits, trims angle of attack and elevator, and propagates airspeed using realized acceleration. This separation is important: jerk minimization reduces artificial command discontinuities, while force-balance logic makes the reference compatible with the physical model.

The contributions of the paper are as follows. First, a six-degree-of-freedom VTOL plant model is derived using a direction cosine matrix attitude representation and Kane's multi-body equations. Second, the model incorporates two tilting rotors with thrust lag, tilt-rate limits, rotor reaction torque, momentum-theory induced velocity, cross-flow attenuation, thrust-inflow correction, and ground effect. Third, the fixed-wing aerodynamic model includes prop wash-modified local dynamic pressure, forward-flow effectiveness, stall blending, and conventional stability-derivative force and moment coefficients. Fourth, a smooth transition reference generator is proposed that uses minimum-jerk phase schedules, thrust-limited force balance, trim angle of attack, trim elevator, coordinated-turn bank, and actuator-aware smoothing. The resulting framework is designed for closed-loop control development and comparative simulation of hover-to-cruise and cruise-to-hover transitions.

III. Literature Review and Modelling Rationale

A. Transition-flight VTOL modeling

Tilt-rotor and tilt-wing aircraft combine the runway independence of rotor-craft with the forward speed and range advantages of fixed-wing aircraft. Historical NASA work on the XV–15 program describes this class of aircraft as a means of combining helicopter-like vertical takeoff, landing, and hover with turboprop-like cruise efficiency [1]. This dual operating envelope makes the transition regime fundamentally more complex than either pure hover or pure cruise. During conversion, the vehicle must continuously redistribute vertical support, forward acceleration, attitude control, and disturbance rejection among rotor thrust, rotor tilt angle, aerodynamic lift, and control-surface moments. Recent transition-control work similarly emphasizes that tilt-rotor VTOL vehicles are attractive because of redundant control authority, but that this same redundancy creates a mode-transition problem in which controller switching, control allocation, actuator saturation, and trajectory smoothness must be handled carefully [2, 3].

The literature contains a spectrum of modelling approaches. At the high-fidelity end, rotor-craft and tilt-rotor prediction may use blade-element, free-wake, vortex-lattice, or computational-fluid-dynamics methods to resolve rotor/wing interaction, wake contraction, and unsteady inflow. These models are appropriate for detailed aerodynamic prediction but are often too expensive or too highly parameterized for real-time control and repeated trajectory generation. At the control-oriented end, many flight-dynamics models represent the airframe as a rigid body and model rotors as thrust-producing actuators applied at specified points. NASA's generic multi-body tilt-rotor modelling work notes that many simulations approximate the vehicle as a single rigid body with rotors represented as thrust application points, a simplification that is useful when appendage mass and relative motion are secondary to the dominant rigid-body dynamics [6]. The model used in this work follows a control-oriented philosophy: it retains the dominant force and moment mechanisms needed for transition analysis but avoids a full-blade-resolved or free-wake aerodynamic model.

A key difficulty in transition flight is that the rotor wake interacts with the wing, fuselage, and control surfaces. NASA rotor/wing studies show that rotor-on-wing and wing-on-rotor interactions are already important in hover and low-speed operation, where the rotor wake impinges on lifting surfaces and modifies both download and effective local flow [5]. These effects are particularly relevant for transitional aircraft, because the control surfaces may receive little free-stream dynamic pressure in hover but may still experience rotor slipstream. A reduced-order model should therefore distinguish the free-stream air-relative velocity from the local velocity experienced by surfaces inside the rotor wake. The present model does this by computing an air-relative body velocity and then adding a weighted induced-velocity increment to the local dynamic pressure used by the elevator, aileron, and rudder.

B. Rigid-body and multi-body dynamics formulation

The vehicle equations of motion are derived as a six-degree-of-freedom rigid-body model augmented by rotor tilt and thrust states and a low-order internal slosh mode. A direction-cosine-matrix attitude representation is used rather than Euler angles because transition maneuvers can involve large pitch excursions and because Euler parameterizations introduce coordinate singularities. With $\mathbf{R}_{bn} \in SO(3)$ denoting the map from body components to inertial components, the attitude kinematics are written

$$\dot{\mathbf{R}}_{bn} = \mathbf{R}_{bn}(\omega_b)_\times, \tag{1}$$

where $\omega_{b\times}$ is the skew-symmetric matrix associated with body angular velocity $\omega_b = [p, q, r]^T$. This form keeps the rotational kinematics directly on $SO(3)$, up to numerical projection or normalization during integration.

Kane's method is well-suited to this model because it provides a systematic procedure for assembling coupled equations of motion from generalized coordinates, generalized speeds, generalized active forces, and generalized inertia forces [7, 8]. Compared with manually expanding Newton–Euler equations, Kane's method is particularly convenient when the system includes non-conservative loads, moving internal masses, and forces applied at offset points. The resulting system can be written in mass-matrix form,

$$M(q,t)\dot{u} = f(q,u,t), \tag{2}$$

where q are generalized coordinates and u are generalized speeds. This form is directly usable in numerical simulation, nonlinear model-predictive control, and trajectory feasibility analysis. The symbolic and numerical implementation is also consistent with modern scientific-computing practice in which computer algebra is used to form and lambdify multi-body equations for repeated simulation [13].

C. Rotor induced flow and slipstream modelling

Rotor-induced flow is commonly approximated at first order using actuator-disk momentum theory. In hover, equating rotor thrust to the change in axial momentum gives

$$v_{i,h} = \sqrt{\frac{T_i}{2\rho A_i}}, \tag{3}$$

where T_i is rotor thrust, ρ is air density, and A_i is rotor disk area. Momentum theory and actuator-disk models are standard first approximations in rotor-craft analysis because they connect thrust, disk loading, and induced velocity without introducing blade flapping, blade-element, or wake-state variables [4, 14]. For a control-oriented VTOL model, Eq. 3 is attractive because it is algebraic, transparent, and computationally inexpensive.

However, actuator-disk hover theory alone is not sufficient to represent transition flight. In forward or cross-flowing conditions, the induced-velocity distribution becomes nonuniform and unsteady. Dynamic-inflow models such as the Pitt–Peters model were introduced to predict low-frequency inflow dynamics and are widely used in rotor-craft stability, control, and real-time simulation [15, 16]. Finite-state inflow models further generalize this idea by representing the induced-flow field with a finite number of aerodynamic states, enabling coupling with rotor-craft flight dynamics without resolving a full wake [17]. These models provide higher fidelity than a static actuator-disk approximation, but they also introduce additional states and parameters and may be difficult to identify for a novel two-tilting-rotor configuration.

The present model, therefore, uses a deliberately reduced-order induced-flow treatment. The hover value in Eq. (3) is retained as the baseline induced velocity, while crossflow attenuation is introduced to decrease coherent slipstream contribution as the air-relative velocity becomes increasingly perpendicular to the rotor thrust axis:

$$v_{i,\text{eff}} = \frac{v_{i,h}}{\sqrt{1 + \left(k_{\text{cf}} V_{\perp,i} / \max(v_{i,h}, \epsilon)\right)^2}}. \tag{4}$$

Here $V_{\perp,i}$ is the cross flow component relative to rotor i, k_{cf} is a tunable attenuation coefficient, and ϵ prevents division by zero. Eq. 4 should not be interpreted as a substitute for dynamic inflow; rather, it is a low-order surrogate that preserves the physically important trend that coherent slipstream impingement on the airframe decreases as the rotor wake is swept or distorted by crossflow. This is a reasonable compromise for trajectory generation and controller development, provided that the coefficient is calibrated or sensitivity-tested.

D. Ground effect and thrust-inflow correction

Rotor performance near the ground differs from out-of-ground-effect performance because the ground plane alters the induced velocity field and reduces the power required to produce a given thrust. Classical work by Cheeseman and Bennett developed an approximate ground-effect treatment for helicopter rotors in forward flight [18]. Later rotorcraft literature has continued to model ground effect using height-dependent or image-system approximations for real-time simulation and control-oriented analysis [19]. For the present VTOL model, a bounded height-dependent multiplier is used,

$$G_i(h_i) = 1 + \frac{G_{\max}}{1 + (h_i/R_i)^n}, \tag{5}$$

where h_i is rotor height above ground, $R_i = \sqrt{A_i/\pi}$ is rotor radius, $G_{\max}$ is the maximum gain, and n controls decay with height. This form is not intended to reproduce detailed ground-vortex or wall-jet structures; it captures the first-order increase in rotor effectiveness near the ground while remaining bounded and differentiable enough for simulation.

The thrust applied by a propulsor also changes with inflow. Momentum theory predicts that induced velocity, propulsive efficiency, and thrust response depend on axial and oblique inflow, and dynamic-inflow literature shows that these effects are state-dependent in maneuvering flight [4, 15]. Because the present model does not include blade-element states or rotor-speed dynamics, inflow dependence is represented by an algebraic thrust-lapse factor,

$$\eta_i = \text{clip}\left[\left(1 + k_a\left(\frac{|V_{a,i}|}{V_{\text{ref}}}\right)^2 + k_c\left(\frac{V_{\perp,i}}{V_{\text{ref}}}\right)^2\right)^{-1/2}, \eta_{\min}, 1\right], \tag{6}$$

where $V_{a,i}$ is axial inflow, $V_{\perp,i}$ is crossflow, and $\eta_{\min}$ prevents unrealistically large thrust loss. The effective rotor thrust is then

$$T_{i,\text{eff}} = T_i\, G_i(h_i)\, \eta_i. \tag{7}$$

This structure makes the modeling assumption explicit: the thrust state T_i represents actuator output, while $T_{i,\text{eff}}$ represents the thrust actually applied to the airframe after low-order environmental and inflow corrections.

E. Fixed-wing aerodynamic model and propwash-modified control authority

In the wing-borne portion of the envelope, conventional aircraft flight-dynamics models express forces and moments through nondimensional aerodynamic coefficients and stability/control derivatives. Standard references formulate the aerodynamic model in terms of angle of attack, sideslip, nondimensional angular rates, control deflections, dynamic pressure, reference area, span, and mean aerodynamic chord [20–22]. This approach is appropriate for control-oriented simulation because it separates geometry and flight condition from nondimensional derivative data. A representative longitudinal lift model is

$$C_L = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\frac{q\bar{c}}{2V} + C_{L\delta_e}\delta_e, \tag{8}$$

with analogous derivative expansions for side force, drag, roll moment, pitch moment, and yaw moment.

The difficulty for the transition flight is that a conventional fixed-wing derivative model is not valid in pure hover or broadside flow. Therefore, the present model reduces fixed-wing coefficient authority when the flow is not primarily aligned with the body-forward direction. It also blends attached-flow lift and drag into a post-stall surrogate at large angles of attack relative to the forward flow. This treatment is consistent with the model's role: it should not predict detailed separated-flow aerodynamics, but it should prevent an attached-flow derivative model from producing unrealistic lift and control authority outside its intended regime.

Prop wash effects are handled by modifying local surface dynamic pressure. The rotor slipstream contribution is computed from the induced-velocity model and weighted separately for the elevator, aileron, and rudder. This is motivated by rotor/wing interaction studies showing that wake impingement changes local aerodynamic loading [5]. It also matches the physical behavior of VTOL transition aircraft: in hover, conventional surfaces may be ineffective in the free stream but can regain partial authority when immersed in rotor wake. Thus the aerodynamic model uses the free-stream air-relative velocity for the global airframe force and uses prop wash-modified local dynamic pressure for control-surface increments.

F. Actuator dynamics, saturation, and tilt-thrust coupling

Transition trajectories can be made dynamically infeasible if the model assumes instantaneous thrust, instantaneous rotor tilt, or unlimited control-surface rates. Tilt-rotor simulation studies commonly include actuator bandwidth, position limits, and rate limits. For example, a large-civil-tilt-rotor simulation work modelled swash plate actuators with rate limiting and nacelle actuators as second-order systems with angular position and rate limits [23]. High-speed rotor-craft handling-quality simulations similarly include rotor and aero-surface actuators represented as second-order systems with position and rate limits [24]. Multi-rotor model-identification studies further show that motor/rotor lag, rotor inertia, and inflow corrections can materially improve model fidelity relative to instantaneous thrust assumptions [25].

Accordingly, the propulsion model uses a first-order thrust lag,

$$\dot{T}_i = \frac{T_{i,c} - T_i}{\tau_T}, \tag{9}$$

where $T_{i,c}$ is commanded thrust and τ_T is the thrust time constant. Rotor tilt commands are passed through a finite-bandwidth, rate-limited actuator model rather than being applied instantaneously. A generic second-order representation often used in flight simulation is

$$\ddot{\beta}_i + 2\zeta_\beta \omega_\beta \dot{\beta}_i + \omega_\beta^2 \beta_i = \omega_\beta^2 \beta_{i,c}, \tag{10}$$

with saturation on β_i and $\dot{\beta}_i$. The implementation in this work uses the same principle—finite tilt bandwidth and explicit rate limiting—while keeping the actuator order low enough for real-time integration and control design.

The rotor force and moment contribution for rotor i is written

$$\hat{\boldsymbol{t}}_i^{\mathcal{B}} = [\cos\beta_i,\ 0,\ \sin\beta_i]^T, \tag{11}$$

$$\boldsymbol{F}_i^{\mathcal{B}} = T_{i,\mathrm{eff}} \hat{\boldsymbol{t}}_i^{\mathcal{B}}, \tag{12}$$

$$\boldsymbol{M}_i^{\mathcal{B}} = \boldsymbol{r}_i^{\mathcal{B}} \times \boldsymbol{F}_i^{\mathcal{B}} + \sigma_i Q_i \hat{\boldsymbol{t}}_i^{\mathcal{B}}, \tag{13}$$

where $\boldsymbol{r}_i^{\mathcal{B}}$ is the rotor position relative to the center of mass, Q_i is reaction torque magnitude, and $\sigma_i \in \{+1, -1\}$ is the spin direction. In the present reduced-order model, reaction torque is represented as $Q_i = c_{Q,i} T_i$. This algebraic relation is simpler than a full rotor-speed/power model but preserves the yawing-moment effect of counter-rotating propulsors.

G. Smooth transition references and jerk minimization

The reference trajectory must be smooth enough for a finite-bandwidth aircraft to track. Smooth polynomial trajectory generation is common in aerial robotics because discontinuities in position derivatives translate into aggressive or infeasible thrust, attitude, and actuator commands. For quadrotors, minimum-snap trajectory generation has become a standard approach because snap is directly related to rotor thrust and attitude-rate requirements under differential-flatness mappings [10, 11]. More recent VTOL fixed-wing work uses differential flatness and snap minimization to generate aggressive trajectories over a broad flight envelope, including post-stall and aerobatic maneuvers [12]. Hybrid-UAV guidance literature has also emphasized jerk-level dynamic inversion as a way to track position references accurately while respecting the dynamics of vehicles with mixed aerodynamic and propulsive control authority [26].

For the present transition reference, the objective is not aggressive aerobatic flight but a smooth, actuator-trackable conversion between hover and cruise. Therefore, a quintic minimum-jerk blend is used for scheduled variables such as altitude, rotor tilt, speed, and heading:

$$\sigma(s) = 10s^3 - 15s^4 + 6s^5, \qquad s \in [0, 1]. \tag{14}$$

This polynomial satisfies

$$\sigma(0) = 0, \quad \sigma(1) = 1, \quad \sigma'(0) = \sigma'(1) = 0, \quad \sigma''(0) = \sigma''(1) = 0, \tag{15}$$

so the scheduled variable has zero endpoint velocity and acceleration for a rest-to-rest segment. The minimum-jerk principle was originally proposed as an experimentally supported model for smooth human reaching motions, where the cost is the integral of squared jerk [9]. In robotics and guidance, the same polynomial is widely used as a closed-form way to obtain smooth finite-duration transitions with matched endpoint derivatives.

Minimum jerk alone does not guarantee dynamic feasibility for a VTOL aircraft. A smooth kinematic schedule could still demand more thrust, tilt rate, or control-surface authority than the plant can supply. Therefore, the present reference generator uses minimum-jerk only to impose smooth phase transitions, then enforces feasibility via force-balance logic. At each point in the transition, the generator estimates lift and drag, applies thrust saturation, imposes a feasible tilt floor for vertical support, computes forward thrust demand, and trims angle of attack and elevator deflection. This division of responsibility is important: jerk minimization reduces command discontinuities and actuator excitation, while the thrust and aerodynamic balance calculations ensure that the resulting reference remains compatible with the physical model.

H. Positioning of the present model

The proposed VTOL model is best described as a reduced-order, control-oriented transition-flight model. It is more detailed than a pure point-mass or quasi-steady fixed-wing model because it includes a full rigid-body attitude representation, offset tilting rotors, rotor reaction torques, thrust lag, tilt-rate limits, propwash-modified local dynamic pressure, inflow-dependent thrust correction, ground effect, and an internal slosh mode. It is less detailed than a blade-element/free-wake/CFD model because it does not resolve blade loads, wake roll-up, vortex-ring state, or fully unsteady induced-flow distributions. This middle ground is consistent with the literature: high-fidelity rotor-craft aerodynamics are necessary for final performance prediction, but lower-order models remain essential for simulation, reference generation, controller development, and sensitivity analysis [4, 6, 25].

The modelling assumptions are therefore justified as follows. First, Kane's method provides a systematic and computationally convenient way to derive the coupled rigid-body/slosh equations. Second, actuator-disk momentum theory gives a transparent baseline induced-velocity model, while cross-flow attenuation and thrust-lapse factors add the leading transition-flight trends without introducing poorly identified wake states. Third, a coefficient-based fixed-wing aerodynamic model is standard for control-oriented aircraft simulation, and its authority is intentionally reduced outside forward-flow conditions. Fourth, actuator lags and rate limits are included to prevent transition commands from being unrealistically abrupt. Finally, minimum-jerk scheduling is justified because the reference should be smooth at phase boundaries, while dynamic feasibility is supplied separately by force-balance and saturation checks.

IV. VTOL Dynamic Model Derivation

The Vertical Takeoff and Landing Fixed-wing model is presented in Figure 1; the initial design is chosen as a representative fixed-wing aircraft. With the defined characteristics for the aircraft being found in Table 3 below.

A. Coordinate Frames and State Definition

Let $\mathcal{N}$ denote the inertial frame and $\mathcal{B}$ denote the body-fixed frame attached to the vehicle center of mass. The direction cosine matrix $R_{bn} \in SO(3)$ maps vectors from body coordinates to inertial coordinates such that

$$v_N = R_{bn} v_B, \qquad v_B = R_{bn}^T v_N. \tag{16}$$

The body angular velocity is expressed in body coordinates as

$$\omega_B = \begin{bmatrix} p & q & r \end{bmatrix}^T. \tag{17}$$

The generalized coordinates are defined as

$$q_g = \begin{bmatrix} r_N^T & \mathrm{vec}(R_{bn})^T & s^T & \beta_1 & \beta_2 \end{bmatrix}^T, \tag{18}$$

where

$$r_N = \begin{bmatrix} x & y & z \end{bmatrix}^T, \qquad s = \begin{bmatrix} s_x & s_y & s_z \end{bmatrix}^T. \tag{19}$$

Table 2 Literature-supported modeling choices used in the VTOL transition model.

Modeling choice	Rationale	Representative literature
Rigid body with thrust-point rotors	Retains dominant vehicle-level force and moment coupling while avoiding blade-resolved dynamics.	Pei and Roithmayr [6]; Maisel et al. [1]
Kane-method multi-body derivation	Efficiently handles generalized speeds, moving internal masses, offset loads, and non-conservative forces.	Kane and Levinson [7]; Lesser [8]
Momentum-theory induced flow	Provides algebraic disk-loading relation for control-oriented hover/low-speed modelling.	Johnson [14]; Leishman [4]
Cross-flow and thrust-lapse corrections	Adds transition-flight wake/inflow trends without full dynamic-inflow states.	Pitt and Peters [15]; Peters et al. [17]
Ground-effect gain	Captures first-order near-ground thrust effectiveness changes.	Cheeseman and Bennett [18]; Khromov and Rand [19]
Coefficient-based fixed-wing aerodynamics	Standard structure for flight-dynamics simulation and control design.	Etkin and Reid [20]; Stevens et al. [21]
Actuator lags and rate limits	Prevents unrealistic instantaneous thrust, nacelle/tilt, and surface commands.	Lawrence et al. [23]; Berger et al. [24]; Niemiec and Gandhi [25]
Minimum-jerk schedules	Ensures smooth endpoint velocity and acceleration for transition phases.	Flash and Hogan [9]; Mellinger and Kumar [10]; Rohr et al. [26]

Here s denotes the internal slosh displacement in body coordinates, and β_1, β_2 are the two rotor tilt angles.

The generalized speed vector is

$$\nu = \begin{bmatrix} v_N^T & \omega_B^T & \dot{s}^T & \dot{\beta}_1 & \dot{\beta}_2 \end{bmatrix}^T. \tag{20}$$

The complete state vector is therefore

$$x = \begin{bmatrix} q_g^T & \nu^T & T_1 & T_2 \end{bmatrix}^T \in \mathbb{R}^{30}. \tag{21}$$

B. Kinematics on $SO(3)$

The translational kinematics are

$$\dot{r}_N = v_N. \tag{22}$$

Since R_{bn} maps body coordinates to inertial coordinates and the angular velocity is expressed in the body frame, the attitude kinematics are

$$\dot{R}_{bn} = R_{bn}\hat{\omega}_B, \tag{23}$$

where

$$\hat{\omega}_B = \begin{bmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{bmatrix}. \tag{24}$$

The rotation matrix satisfies the $SO(3)$ constraints

$$R_{bn}^T R_{bn} = I, \qquad \det(R_{bn}) = 1. \tag{25}$$

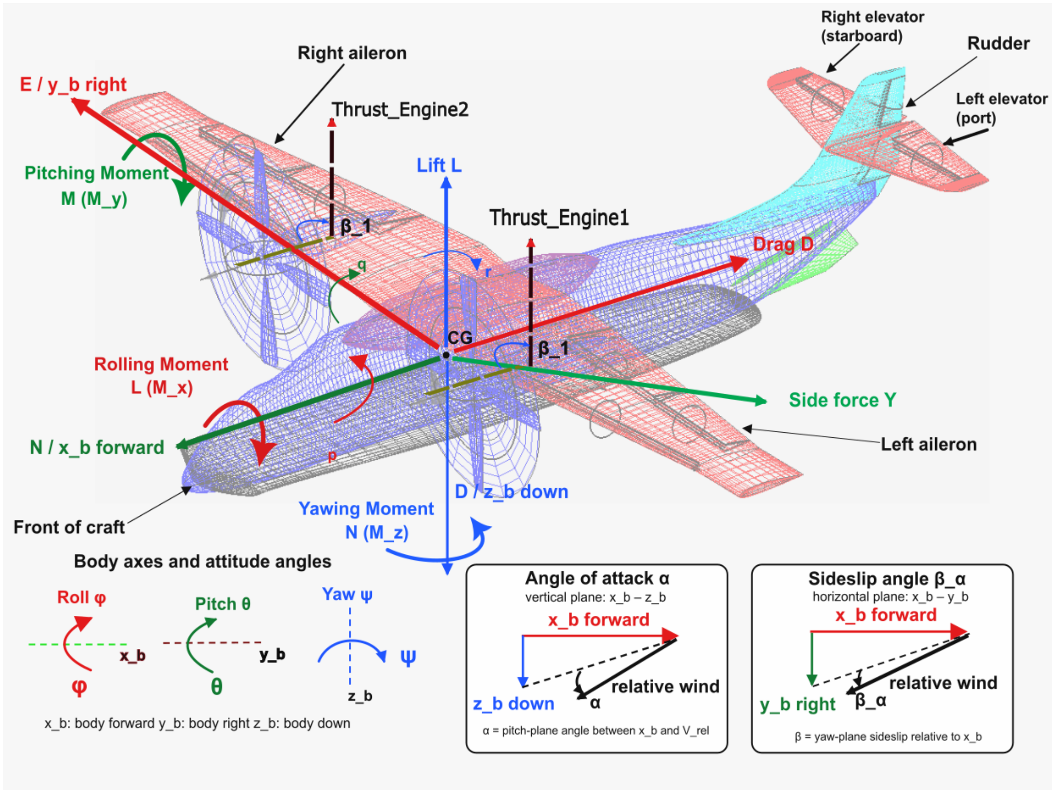

Fig. 1 WildFire Fighting VTOL Aircraft Model and Force Definition

C. Rotor Force and Moment Model

The aircraft contains two tilting rotors mounted symmetrically about the body x–z plane. Their position vectors relative to the center of mass are

$$r_1 = \begin{bmatrix} x_a \\ y_a \\ z_a \end{bmatrix}, \qquad r_2 = \begin{bmatrix} x_a \\ -y_a \\ z_a \end{bmatrix}. \tag{26}$$

Each rotor tilts in the body x-z plane. The thrust direction of rotor i is

$$t_i(\beta_i) = \begin{bmatrix} \cos\beta_i \\ 0 \\ \sin\beta_i \end{bmatrix}. \tag{27}$$

Thus, the thrust force produced by rotor i is

$$F_{T_i,B} = T_i t_i(\beta_i). \tag{28}$$

The total rotor force in the body frame is

$$F_{T,B} = T_1 t_1(\beta_1) + T_2 t_2(\beta_2). \tag{29}$$

The thrust-induced moment about the center of mass is

$$\tau_{T,B} = r_1 \times T_1 t_1(\beta_1) + r_2 \times T_2 t_2(\beta_2). \tag{30}$$

Each rotor also produces a reaction torque proportional to thrust:

$$Q_i = c_{Q_i} T_i. \tag{31}$$

Table 3 Representative VTOL aircraft parameters used in the full-plant replay model

Parameter	Symbol	Value
Mass	m	1325 kg
Slosh modal mass	m_s	$0.3m = 397.5$ kg
Wing area	S	21.667 m^2
Wing span	b	14.719 m
Mean aerodynamic chord	$\bar{c}$	1.472 m
Rotor lateral arm	y_a	3.0 m
Rotor vertical offset	z_a	0.5 m
Maximum rotor thrust	$T_{\max}$	12000 N
Rotor tilt range	β	$[0, 90°]$
Rotor tilt-rate cap	$\dot{\beta}_{\max}$	$40°/\mathrm{s}$
Thrust actuator time constant	τ_T	0.15 s
Tilt actuator time constant	τ_β	0.20 s

Therefore, the total rotor reaction torque is

$$\tau_{Q,B} = \sigma_1 c_{Q_1} T_1 t_1(\beta_1) + \sigma_2 c_{Q_2} T_2 t_2(\beta_2), \tag{32}$$

where $\sigma_i \in \{-1, 1\}$ denotes the rotor spin direction.

D. Aerodynamic Force and Moment Model

The body-frame velocity is

$$v_B = R_{bn}^\top v_N = \begin{bmatrix} u_b & v_b & w_b \end{bmatrix}^\top. \tag{33}$$

The airspeed, angle of attack, sideslip angle, and dynamic pressure are

$$V = \sqrt{u_b^2 + v_b^2 + w_b^2}, \tag{34}$$

$$\alpha = \tan^{-1}\left(\frac{w_b}{u_b}\right), \tag{35}$$

$$\beta_a = \sin^{-1}\left(\frac{v_b}{V}\right), \tag{36}$$

$$\bar{q} = \frac{1}{2}\rho V^2. \tag{37}$$

The nondimensional angular rates are

$$\hat{p} = \frac{pb}{2V}, \qquad \hat{q} = \frac{q\bar{c}}{2V}, \qquad \hat{r} = \frac{rb}{2V}. \tag{38}$$

The aerodynamic coefficients are modelled as

$$C_L = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\hat{q} + C_{L\delta_e}\delta_e, \tag{39}$$

$$C_D = C_{D0} + k_D C_L^2, \tag{40}$$

$$C_Y = C_{Y\beta}\beta_a + C_{Y\delta_r}\delta_r, \tag{41}$$

$$C_\ell = C_{\ell\beta}\beta_a + C_{\ell p}\hat{p} + C_{\ell r}\hat{r} + C_{\ell\delta_a}\delta_a + C_{\ell\delta_r}\delta_r, \tag{42}$$

$$C_m = C_{m0} + C_{m\alpha}\alpha + C_{mq}\hat{q} + C_{m\delta_e}\delta_e, \tag{43}$$

$$C_n = C_{n\beta}\beta_a + C_{np}\hat{p} + C_{nr}\hat{r} + C_{n\delta_a}\delta_a + C_{n\delta_r}\delta_r. \tag{44}$$

The aerodynamic force in the wind axes is

$$F_w = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} = \begin{bmatrix} -\bar{q}SC_D \\ \bar{q}SC_Y \\ -\bar{q}SC_L \end{bmatrix}. \tag{45}$$

Transforming this force into the body frame gives

$$F_{a,B} = R_y(\alpha)R_z(\beta_a)F_w. \tag{46}$$

The aerodynamic moment in the body frame is

$$\tau_{a,B} = \begin{bmatrix} \bar{q}SbC_\ell \\ \bar{q}S\bar{c}C_m \\ \bar{q}SbC_n \end{bmatrix}. \tag{47}$$

E. Slosh Dynamics

The slosh mass is modelled as a three-degree-of-freedom internal particle with displacement s relative to the body frame. The restoring and damping force acting on the slosh mass is

$$F_{s,B} = \begin{bmatrix} -k_x s_x - c_x \dot{s}_x \\ -k_y s_y - c_y \dot{s}_y \\ -k_z s_z - c_z \dot{s}_z \end{bmatrix}. \tag{48}$$

The equal and opposite force acts on the rigid body, producing coupling between the internal fluid motion and the aircraft's rigid-body dynamics.

F. Kane's Equations

The total aircraft mass is decomposed into a rigid-body contribution $m - m_s$ and a slosh modal mass m_s, while preserving the total vehicle mass m. Kane's method is used to obtain the equations of motion in the form

$$M(q_g)\dot{\nu} = F(q_g, \nu, T_1, T_2, \beta_1, \beta_2, \delta_e, \delta_a, \delta_r), \tag{49}$$

where $M(q_g)$ is the generalized mass matrix and F contains gravity, aerodynamic forces and moments, rotor thrust forces, rotor reaction torques, and slosh coupling forces.

The resulting continuous-time system can be written as

$$\dot{x} = \begin{bmatrix} \dot{q}_g \\ \dot{\nu} \\ \dot{T}_1 \\ \dot{T}_2 \end{bmatrix} = \begin{bmatrix} \Gamma(q_g, \nu) \\ M(q_g)^{-1}F(q_g, \nu, T_1, T_2, \beta_1, \beta_2, \delta) \\ \dot{T}_1 \\ \dot{T}_2 \end{bmatrix}, \tag{50}$$

where

$$\delta = \begin{bmatrix} \delta_e & \delta_a & \delta_r \end{bmatrix}^T.$$

G. Control Surface Effectiveness and Local Flow Modeling

In conventional fixed-wing models, aerodynamic forces and moments are expressed using the freestream dynamic pressure

$$\bar{q} = \frac{1}{2}\rho V^2, \tag{51}$$

where $V = \|v_B\|$ is the airspeed at the center of mass. However, for VTOL configurations, particularly during hover and transition, this assumption becomes invalid. In these regimes, the freestream velocity may be small while the control surfaces remain immersed in rotor-induced flow.

To account for this effect, the aerodynamic control effectiveness is modelled using a *local effective velocity* at each control surface. For a control surface $j \in \{e, a, r\}$ (elevator, aileron, rudder), the local body-frame velocity is defined as

$$v_{B,j} = v_B + \omega_B \times r_j + v_{\mathrm{prop},j}, \tag{52}$$

where r_j is the position vector from the center of mass to the control surface, and $v_{\mathrm{prop},j}$ represents the rotor-induced slipstream velocity at that location.

The corresponding effective airspeed is

$$V_{\mathrm{eff},j} = \|v_{B,j}\|, \tag{53}$$

and the local dynamic pressure becomes

$$\bar{q}_j = \frac{1}{2}\rho V_{\mathrm{eff},j}^2. \tag{54}$$

1. Rotor-Induced Velocity Model

The rotor-induced velocity is modelled using a momentum-theory approximation. For a two-rotor configuration, the induced velocity contribution at surface j is

$$v_{\mathrm{prop},j} = \sum_{i=1}^{2} \eta_{ij} \sqrt{\frac{T_i}{2\rho A_i}}\, t_i(\beta_i), \tag{55}$$

where T_i is the thrust generated by rotor i, A_i is the rotor disk area, $t_i(\beta_i)$ is the rotor thrust direction, and $\eta_{ij} \in [0, 1]$ is a weighting factor representing the fraction of slipstream from rotor i that impinges on surface j.

2. Modified Aerodynamic Control Effectiveness

Using the local dynamic pressure, the control-surface-induced force and moment increments are written as

$$\Delta F_{a,B} = R_y(\alpha) R_z(\beta_a) \begin{bmatrix} 0 \\ \bar{q}_r S C_{Y\delta_r} \delta_r \\ -\bar{q}_e S C_{L\delta_e} \delta_e \end{bmatrix}, \tag{56}$$

$$\Delta \tau_{a,B} = \begin{bmatrix} \bar{q}_a S b C_{\ell\delta_a} \delta_a \\ \bar{q}_e S \bar{c} C_{m\delta_e} \delta_e \\ \bar{q}_r S b C_{n\delta_r} \delta_r \end{bmatrix}. \tag{57}$$

This formulation ensures that aerodynamic control authority remains non-zero in low-speed regimes where rotor-induced flow dominates. In particular, during hover and transition flight ($V \approx 0$), the control surfaces can still generate significant forces and moments due to the slipstream velocity.

The resulting model captures the coupling between propulsion and aerodynamic control, which is critical for accurately representing VTOL dynamics across the full flight envelope.

H. Thrusting Actuator Dynamics

The rotor thrust actuators are modelled as first-order systems:

$$\dot{T}_1 = \frac{T_{\mathrm{cmd},1} - T_1}{\tau_T}, \tag{58}$$

$$\dot{T}_2 = \frac{T_{\mathrm{cmd},2} - T_2}{\tau_T}. \tag{59}$$

The rotor tilt actuators are modelled as second-order systems:

$$\ddot{\beta}_1 = -\frac{1}{\tau_\beta}\dot{\beta}_1 - \frac{1}{\tau_\beta^2}\beta_1 + \frac{1}{\tau_\beta^2}\beta_{\mathrm{cmd},1}, \tag{60}$$

$$\ddot{\beta}_2 = -\frac{1}{\tau_\beta}\dot{\beta}_2 - \frac{1}{\tau_\beta^2}\beta_2 + \frac{1}{\tau_\beta^2}\beta_{\mathrm{cmd},2}. \tag{61}$$

I. Control-Affine Form

For control design, the system is written in the nonlinear control-affine form

$$\dot{x} = f(x) + G(x)u + d(x,t), \tag{62}$$

where $d(x,t)$ represents lumped model uncertainty, wind disturbance, aerodynamic mismatch, actuator saturation effects, and unmodelled slosh effects.

The input vector is chosen as

$$u = \begin{bmatrix} T_{\text{cmd},1} & T_{\text{cmd},2} & \beta_{\text{cmd},1} & \beta_{\text{cmd},2} & \delta_e & \delta_a & \delta_r \end{bmatrix}^T. \tag{63}$$

The state is

$$x = \begin{bmatrix} q_g^T & \nu^T & T_1 & T_2 \end{bmatrix}^T \in \mathbb{R}^{30}, \tag{64}$$

with

$$q_g = \begin{bmatrix} r_N^T & \text{vec}(R_{bn})^T & s^T & \beta_1 & \beta_2 \end{bmatrix}^T \in \mathbb{R}^{17}, \tag{65}$$

and

$$\nu = \begin{bmatrix} v_N^T & \omega_B^T & \dot{s}^T & \dot{\beta}_1 & \dot{\beta}_2 \end{bmatrix}^T \in \mathbb{R}^{11}. \tag{66}$$

The kinematic mapping $\Gamma(q_g,\nu)$ is defined by

$$\dot{q}_g = \Gamma(q_g,\nu), \tag{67}$$

where

$$\Gamma(q_g,\nu) = \begin{bmatrix} v_N \\ \text{vec}\left(R_{bn}\widehat{\omega}_B\right) \\ \dot{s} \\ \dot{\beta}_1 \\ \dot{\beta}_2 \end{bmatrix} \in \mathbb{R}^{17}. \tag{68}$$

The generalized speed dynamics are obtained from Kane's equations. In compact form,

$$M(q_g)\dot{\nu} = F(q_g,\nu,T_1,T_2,\beta_1,\beta_2,\delta), \tag{69}$$

where $M(q_g) \in \mathbb{R}^{11\times 11}$ is the generalized mass matrix associated with the speed vector ν. This matrix contains the rigid-body inertia, translational mass terms, slosh modal mass coupling, and the coupling induced by the generalized coordinates. In the numerical implementation, this matrix is generated symbolically using Kane's method and evaluated together with the forcing vector.

For the control-affine decomposition, the forcing term is separated as

$$F(q_g,\nu,T_1,T_2,\beta_1,\beta_2,\delta) = F_0(x) + B_\delta(x)\delta + \Delta_\delta(x,\delta), \tag{70}$$

where

$$\delta = \begin{bmatrix} \delta_e & \delta_a & \delta_r \end{bmatrix}^T. \tag{71}$$

The vector $F_0(x)$ contains all generalized forces that are independent of the current aerodynamic control-surface deflections. Therefore, $F_0(x)$ includes gravity, baseline aerodynamic forces and moments, rotor thrust forces generated by the current thrust states T_1, T_2, rotor reaction torques, slosh restoring and damping forces, and the drift part of the tilt actuator dynamics.

More explicitly, $F_0(x)$ may be written as

$$F_0(x) = \begin{bmatrix} F_{\text{Kane},0}(q_g,\nu,T_1,T_2,\beta_1,\beta_2) \\ -\dfrac{1}{\tau_\beta}\dot{\beta}_1 - \dfrac{1}{\tau_\beta^2}\beta_1 \\ -\dfrac{1}{\tau_\beta}\dot{\beta}_2 - \dfrac{1}{\tau_\beta^2}\beta_2 \end{bmatrix}, \tag{72}$$

where $F_{\text{Kane},0} \in \mathbb{R}^9$ represents the translational, rotational, and slosh generalized forcing terms obtained from Kane's equations with $\delta_e = \delta_a = \delta_r = 0$. The last two rows correspond to the drift part of the second-order tilt actuator dynamics.

The control-surface effectiveness matrix $B_\delta(x) \in \mathbb{R}^{11\times 3}$ maps the elevator, aileron, and rudder deflections into generalized forces. It is obtained by projecting the control-surface aerodynamic wrench into the Kane generalized coordinates:

$$B_\delta(x) = \begin{bmatrix} J_W(q_g)^T \mathcal{B}_\delta(x) \\ 0_{2\times 3} \end{bmatrix}, \tag{73}$$

where $J_W(q_g)^T$ is the Kane generalized-force mapping from body-frame force and moment increments into the first nine generalized speed equations, and

$$\mathcal{B}_\delta(x) = \begin{bmatrix} \mathcal{B}_{F,\delta}(x) \\ \mathcal{B}_{\tau,\delta}(x) \end{bmatrix} \in \mathbb{R}^{6\times 3} \tag{74}$$

is the body-frame aerodynamic wrench effectiveness matrix.

Using the local dynamic pressures $\bar{q}_e, \bar{q}_a, \bar{q}_r$, the force-effectiveness block is

$$\mathcal{B}_{F,\delta}(x) = R_y(\alpha) R_z(\beta_a) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \bar{q}_r S C_{Y\delta_r} \\ -\bar{q}_e S C_{L\delta_e} & 0 & 0 \end{bmatrix}, \tag{75}$$

and the moment-effectiveness block is

$$\mathcal{B}_{\tau,\delta}(x) = \begin{bmatrix} 0 & \bar{q}_a S b C_{\ell\delta_a} & 0 \\ \bar{q}_e S \bar{c} C_{m\delta_e} & 0 & 0 \\ 0 & 0 & \bar{q}_r S b C_{n\delta_r} \end{bmatrix}. \tag{76}$$

The term $\Delta_\delta(x, \delta)$ collects aerodynamic terms that are not exactly affine in δ, such as the quadratic drag dependence

$$C_D = C_{D0} + k_D C_L^2. \tag{77}$$

For the nominal control-affine model, $\Delta_\delta(x, \delta)$ is either neglected or absorbed into $d(x, t)$. Thus,

$$M(q_g)\dot{v} = F_0(x) + B_\delta(x)\delta. \tag{78}$$

The drift vector is therefore

$$f(x) = \begin{bmatrix} \Gamma(q_g, v) \\ M(q_g)^{-1} F_0(x) \\ -\dfrac{1}{\tau_T} T_1 \\ -\dfrac{1}{\tau_T} T_2 \end{bmatrix}. \tag{79}$$

The input matrix is partitioned as

$$G(x) = \begin{bmatrix} G_T(x) & G_\beta(x) & G_\delta(x) \end{bmatrix}. \tag{80}$$

The thrust command mapping is

$$G_T(x) = \begin{bmatrix} 0_{28\times 2} & \\ \dfrac{1}{\tau_T} & 0 \\ 0 & \dfrac{1}{\tau_T} \end{bmatrix}. \tag{81}$$

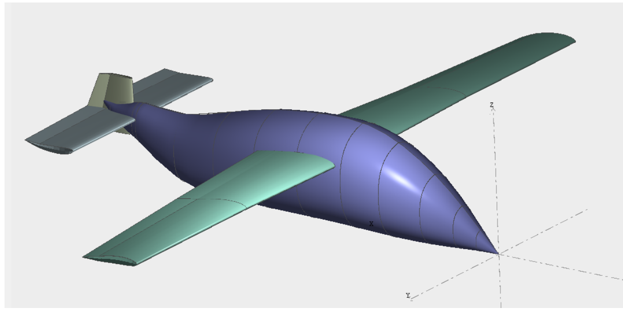

Fig. 2 XFLR5 Aircraft Frame Design

The tilt command mapping is

$$G_\beta(x) = \begin{bmatrix} 0_{26\times 2} & \\ \dfrac{1}{\tau_\beta^2} & 0 \\ 0 & \dfrac{1}{\tau_\beta^2} \\ 0_{2\times 2} & \end{bmatrix}. \tag{82}$$

The aerodynamic control-surface mapping is

$$G_\delta(x) = \begin{bmatrix} 0_{17\times 3} \\ M(q_g)^{-1}B_\delta(x) \\ 0_{2\times 3} \end{bmatrix}. \tag{83}$$

Therefore, the complete control-affine representation is

$$\boxed{\dot{x} = f(x) + G_T(x)\begin{bmatrix} T_{\text{cmd},1} \\ T_{\text{cmd},2} \end{bmatrix} + G_\beta(x)\begin{bmatrix} \beta_{\text{cmd},1} \\ \beta_{\text{cmd},2} \end{bmatrix} + G_\delta(x)\begin{bmatrix} \delta_e \\ \delta_a \\ \delta_r \end{bmatrix} + d(x,t)} \tag{84}$$

V. Aerodynamic Coefficient Generation

To develop the aerodynamic coefficients used in the model, a simplified approach using XFLR5 is adopted. Seen in Fig. 2, the model resembles the given wire frame in Fig. 1.

In the model, baseline vehicle parameters are also used to generate the aerodynamic coefficients; these were derived from empirical relations and primarily serve as estimates of what a physical VTOL fixed-wing platform would be. This information is presented in Table 4 below, which is used to generate the XFLR5 model.

Table 4 Aircraft geometric parameters

Parameter	Symbol	Value	Unit / Notes
Main wing area	S	22.04	m^2
Main wing span	b	14.00	m
Main wing projected span	b_{proj}	14.00	m
Main wing mean geometric chord	$\bar{c}$	1.57	m
Main wing mean aerodynamic chord	MAC	1.59	m
Main wing aspect ratio	AR	8.89	–

Parameter	Symbol	Value	Unit / Notes
Main wing taper ratio	λ	0.68	–
Main wing root-to-tip sweep	Λ	-0.65	deg
Main wing x-location	x_w	3.10	m
Main wing z-location	z_w	1.40	m
Elevator / horizontal tail area	S_h	5.13	m^2
Elevator span	b_h	4.46	m
Elevator mean aerodynamic chord	MAC_h	1.15	m
Elevator aspect ratio	AR_h	3.88	–
Elevator taper ratio	λ_h	0.92	–
Elevator lever arm	l_h	4.74	m
Elevator x-location	x_h	8.00	m
Elevator z-location	z_h	1.30	m
Vertical fin area	S_v	1.04	m^2
Vertical fin span / height	b_v	2.20	m
Vertical fin mean aerodynamic chord	MAC_v	0.97	m
Vertical fin aspect ratio	AR_v	4.63	–
Vertical fin taper ratio	λ_v	0.58	–
Vertical fin root-to-tip sweep	Λ_v	4.42	deg
Vertical fin x-location	x_v	8.20	m
Vertical fin z-location	z_v	0.90	m

Main Wing Definition

The main wing is defined using three spanwise sections. The wing is placed at

$$x_w = 3.10 \text{ m}, \qquad z_w = 1.40 \text{ m}.$$

Table 5 Main wing section definition

Section	y	Chord	Offset	Dihedral	Twist	Airfoil
1	0.000 m	1.850 m	0.000 m	0.0°	0.0°	`NACA 2412`
2	3.500 m	1.600 m	0.070 m	0.0°	5.0°	`NACA 2412 Flap`
3	7.000 m	1.250 m	0.070 m	–	5.0°	`NACA 4412_FLAP`

Elevator Definition

The elevator, or horizontal tail, is defined using two spanwise sections. It is placed at

$$x_h = 8.00 \text{ m}, \qquad z_h = 1.30 \text{ m}.$$

The elevator lever arm is

$$l_h = 4.74 \text{ m}.$$

Vertical Fin Definition

The vertical fin is defined using two spanwise sections. It is placed at

$$x_v = 8.20 \text{ m}, \qquad z_v = 0.90 \text{ m}.$$

Table 6 Elevator section definition

Section	y	Chord	Offset	Dihedral	Twist	Airfoil
1	0.000 m	1.200 m	0.000 m	0.0°	0.0°	`NACA 4412_INV_flap`
2	2.230 m	1.100 m	0.020 m	–	0.0°	`NACA 4412_INV_flap`

Table 7 Vertical fin section definition

Section	y	Chord	Offset	Dihedral	Twist	Airfoil
1	0.000 m	1.200 m	0.000 m	0.0°	0.0°	`NACA 0012_flap`
2	1.100 m	0.700 m	0.040 m	–	0.0°	`NACA 0012_flap`

Derived Parameters

The following useful ratios and distances are obtained from the geometric data.

Table 8 Derived aircraft parameters

Parameter	Formula	Value
Horizontal tail area ratio	$\frac{S_h}{S}$	0.233
Vertical tail area ratio	$\frac{S_v}{S}$	0.047
Horizontal tail volume coefficient	$\frac{S_h l_h}{S \cdot MAC}$	0.694
Coordinate distance between main wing and elevator	$x_h - x_w$	4.90 m
Coordinate distance between main wing and fin	$x_v - x_w$	5.10 m

To develop the aerodynamic coefficients, a weight distribution was added to the XLFR5 system to calculate the aircraft's inertial impact; this also provides an initial indication of the aircraft's inertial mass. The total mass is 1300 kg, with a center of gravity at x = 3.44 m, y = 0, z = 0.618. The Inertia in the body frame is found to be $I_{xx} = 4308.13\frac{kg}{m^2}$, $I_{yy} = 2733.05\frac{kg}{m^2}$, $I_{zz} = 6466.52\frac{kg}{m^2}$ and $I_{xz} = 288.75\frac{kg}{m^2}$.

From this model, dynamic and static coefficients were generated, and the dynamic portions were averaged to find the aerodynamic and control derivatives found in Table 9.

Table 9 Aerodynamic coefficient parameters mapped to the paper notation

Parameter	Paper / model variable	Value	Note
Lift coefficient offset	C_{L0}	0.20	Baseline lift coefficient.
Lift angle-of-attack derivative	$C_{L\alpha}$	5.50	Per radian if α is in radians.
Lift pitch-rate derivative	C_{Lq}	7.00	Multiplies nondimensional pitch rate $\hat{q}$.
Lift elevator derivative	$C_{L\delta_e}$	0.40	Multiplies elevator deflection δ_e.
Maximum lift coefficient	$C_{L,\max}$	1.50	Stall/limiting value used in the plant model.
Zero-lift drag coefficient	C_{D0}	0.04	Baseline drag coefficient.

Parameter	Paper / model variable	Value	Note
Parabolic drag-polar factor	k_D	0.04	Used in $C_D = C_{D0} + k_D C_L^2$.
Side-force sideslip derivative	$C_{Y\beta}$	-0.98	Per radian if β_a is in radians.
Side-force rudder derivative	$C_{Y\delta_r}$	0.17	Multiplies rudder deflection δ_r.
Rolling-moment sideslip derivative	$C_{\ell\beta}$	-0.12	Per radian if β_a is in radians.
Rolling-moment roll-rate derivative	$C_{\ell p}$	-0.50	Multiplies nondimensional roll rate $\hat{p}$.
Rolling-moment yaw-rate derivative	$C_{\ell r}$	0.10	Multiplies nondimensional yaw rate $\hat{r}$.
Rolling-moment aileron derivative	$C_{\ell\delta_a}$	0.08	Multiplies aileron deflection δ_a.
Rolling-moment rudder derivative	$C_{\ell\delta_r}$	0.02	Multiplies rudder deflection δ_r.
Pitching-moment offset	C_{m0}	0.02	Baseline pitching-moment coefficient.
Pitching-moment angle-of-attack derivative	$C_{m\alpha}$	-1.20	Static longitudinal stability derivative.
Pitching-moment pitch-rate derivative	C_{mq}	-12.00	Multiplies nondimensional pitch rate $\hat{q}$.
Pitching-moment elevator derivative	$C_{m\delta_e}$	-1.10	Multiplies elevator deflection δ_e.
Yawing-moment sideslip derivative	$C_{n\beta}$	0.25	Directional stability derivative.
Yawing-moment roll-rate derivative	C_{np}	-0.06	Multiplies nondimensional roll rate $\hat{p}$.
Yawing-moment yaw-rate derivative	C_{nr}	-0.20	Multiplies nondimensional yaw rate $\hat{r}$.
Yawing-moment aileron derivative	$C_{n\delta_a}$	0.02	Multiplies aileron deflection δ_a.
Yawing-moment rudder derivative	$C_{n\delta_r}$	-0.12	Multiplies rudder deflection δ_r.
Sideslip clipping limit	$\beta_{a,\text{lim}}$	20°	Plant-side effective sideslip cap.

VI. Physical Effects on the Aircraft

This section presents a list of parameters that affect the dynamics of the VTOL platform as described in Sec. IV.

Table 10 Current full-plant parameters mapped to the paper variables

Parameter	Paper / model variable	Current value	Note
Mass, atmosphere, and inertia			
Gravitational acceleration	g	9.81 m/s^2	Used in weight and trim calculations.

Parameter	Paper / model variable	Current value	Note
Air density	ρ	$1.225\ \mathrm{kg/m^3}$	Sea-level ISA approximation.
Speed of sound	a	$340.3\ \mathrm{m/s}$	Used if Mach number is evaluated.
Total aircraft mass	m	1300 kg	Total mass including slosh modal mass.
Slosh modal mass	m_s	$0.3m = 397.5$ kg	Internal slosh mass used in the full plant.
Rotor geometry, thrust, and actuator parameters			
Rotor longitudinal offset	x_a	0.0 m	Rotor offset from the center of mass.
Rotor lateral offset	y_a	3.0 m	Rotor 1 uses $+y_a$, rotor 2 uses $-y_a$.
Rotor vertical offset	z_a	0.5 m	Rotor offset from the center of mass.
Rotor 1 position vector	$\mathbf{r}_1$	$[x_a,\ y_a,\ z_a]^T$	Position from CG to rotor 1.
Rotor 2 position vector	$\mathbf{r}_2$	$[x_a,\ -y_a,\ z_a]^T$	Position from CG to rotor 2.
Minimum rotor thrust	$T_{\min}$	0 N	Rotor thrust lower saturation.
Maximum rotor thrust	$T_{\max}$	14000 N	Current code value.
Rotor tilt lower limit	$\beta_{\min}$	0°	Hover/cruise tilt lower bound.
Rotor tilt upper limit	$\beta_{\max}$	90°	Equivalent to $\pi/2$ rad.
Rotor tilt-rate cap	$\dot{\beta}_{\max}$	$40^\circ/\mathrm{s}$	Equivalent to $0.698\ \mathrm{rad/s}$.
Thrust actuator time constant	τ_T	0.15 s	First-order thrust actuator lag.
Rotor-tilt actuator time constant	τ_β	0.20 s	Tilt actuator lag.
Rotor 1 torque-per-thrust coefficient	c_{Q1}	0.02	Reaction torque model $Q_i = c_{Qi}T_i$.
Rotor 2 torque-per-thrust coefficient	c_{Q2}	0.02	Reaction torque model $Q_i = c_{Qi}T_i$.
Rotor 1 spin sign	σ_1	$+1$	Counter-rotating rotor sign.
Rotor 2 spin sign	σ_2	-1	Counter-rotating rotor sign.
Rotor 1 disk area	A_1	$\pi\ \mathrm{m}^2$	Momentum-theory induced velocity.
Rotor 2 disk area	A_2	$\pi\ \mathrm{m}^2$	Momentum-theory induced velocity.
Crossflow attenuation coefficient	k_{cf}	1.0	Propwash coherence attenuation.
Ground-effect maximum gain	$G_{\max}$	0.20	Low-order ground-effect model.
Ground-effect exponent	n	2.0	Ground-effect decay exponent.
Thrust-inflow reference speed	V_{ref}	$15.0\ \mathrm{m/s}$	Reference speed in thrust-lapse model.

Parameter	Paper / model variable	Current value	Note
Axial inflow gain	k_a	0.35	Axial component of thrust-lapse model.
Crossflow inflow gain	k_c	0.20	Crossflow component of thrust-lapse model.
Minimum thrust-inflow factor	$\eta_{\min}$	0.55	Lower bound on thrust-lapse multiplier.
Slosh and tank parameters			
Tank radius	R_{tank}	0.40 m	Tank/slosh geometry bound.
Tank vertical bound	$z_{\text{tank,max}}$	0.25 m	Maximum slosh displacement bound.
Slosh stiffness, x direction	k_x	3000 N/m	Slosh spring coefficient.
Slosh stiffness, y direction	k_y	3000 N/m	Slosh spring coefficient.
Slosh stiffness, z direction	k_z	4000 N/m	Slosh spring coefficient.
Slosh damping, x direction	c_x	600 N s/m	Slosh damping coefficient.
Slosh damping, y direction	c_y	600 N s/m	Slosh damping coefficient.
Slosh damping, z direction	c_z	2500 N s/m	Slosh damping coefficient.

The prop wash portion of the model parameters is given by Table 11.

Table 11 Propwash and control-surface actuator parameters

Parameter	Paper / model variable	Value	Note
Global propwash weights	$(\eta_{1,\text{global}}, \eta_{2,\text{global}})$	(0.3, 0.2)	Rotor slipstream impingement weights.
Elevator propwash weights	(η_{1e}, η_{2e})	(0.2, 0.2)	Used for local elevator dynamic pressure.
Aileron propwash weights	(η_{1a}, η_{2a})	(0.4, 0.4)	Used for local aileron dynamic pressure.
Rudder propwash weights	(η_{1r}, η_{2r})	(0.2, 0.2)	Used for local rudder dynamic pressure.
Elevator actuator time constant	τ_{δ_e}	0.08 s	Plant-side surface actuator lag.
Aileron actuator time constant	τ_{δ_a}	0.08 s	Plant-side surface actuator lag.
Rudder actuator time constant	τ_{δ_r}	0.08 s	Plant-side surface actuator lag.
Elevator rate cap	$\dot{\delta}_{e,\max}$	$60^\circ/\text{s}$	Equivalent to 1.047 rad/s.
Aileron rate cap	$\dot{\delta}_{a,\max}$	$60^\circ/\text{s}$	Equivalent to 1.047 rad/s.
Rudder rate cap	$\dot{\delta}_{r,\max}$	$60^\circ/\text{s}$	Equivalent to 1.047 rad/s.

VII. Reference Trajectory Generation

In this section, the principles applied to develop the tilting-rotor fixed-wing VTOL model are used to derive a feasible full-flight trajectory for a closed-loop feedback control system to track. This tracking problem needs to incorporate a reference trajectory that minimizes the jerk ($\frac{d^3\mathbf{x}}{dt^3}$), i.e., the amount of change in acceleration experienced by the sloshing mass. Therefore, any generated reference must penalize jerk in order to reduce rapid changes in the acceleration demanded by the trajectory. The minimum-jerk principle was originally introduced in [9], which minimized the integral of the squared jerk component. Similar high-order polynomial smoothing ideas are used for other aerial-robotics applications, including minimum-snap quadrotor trajectories [10, 11] and snap-minimizing VTOL fixed-wing trajectory generation for hybrid UAV and eVTOL systems[27]. Jerk-level guidance has also been used for hybrid UAV and eVTOL trajectory tracking methods, where high-order reference information improves compatibility with the vehicle dynamics [28, 29].

In this work, jerk minimization is applied to the scheduled reference variables rather than to the complete full-order state vector. Defining $\xi_r(t)$ to denote a scalar or vector scheduling variance, such as altitude, airspeed, heading, inertial position, or rotor-tilting angles. The smoothness objective for a phase beginning at t_0 and ending at the final t_f is written as

$$J_\xi = \int_{t_0}^{t_f} \left\| \frac{d^3\xi_r(t)}{dt^3} \right\|^2 dt. \tag{85}$$

Because of the defined jerk component earlier,

$$\frac{d^3\xi_r}{dt^3} = \frac{d}{dt}\left(\frac{d^2\xi_r}{dt^2}\right). \tag{86}$$

This minimizes (85), reducing the rapid changes in the commanded acceleration, which is a desirable trait for the present VTOL transition problem because sudden changes in acceleration will change the corresponding force from the internal sloshing payload mass. Reducing jerk will also reduce changes in rotor tilt, attitudes, and control surface effectiveness demands. Reducing the jerking component, therefore, limits the artificial excitation of the vehicle and produces a reference that is more consistent with finite actuator bandwidth.

For each phase, a normalized phase coordinate is introduced,

$$\lambda = \frac{t - t_0}{T_p}, \qquad T_p = t_f - t_0, \qquad \lambda \in [0, 1], \tag{87}$$

where T_p is the phase duration. A quintic minimum-jerk blending polynomial is then used,

$$\sigma(\lambda) = 10\lambda^3 - 15\lambda^4 + 6\lambda^5. \tag{88}$$

The reference value for a scheduled variable is generated by

$$\xi_r(t) = \xi_0 + \left(\xi_f - \xi_0\right)\sigma(\lambda), \tag{89}$$

where ξ_0 and ξ_f are the initial and final values for the phase. The first three derivatives are

$$\dot{\xi}_r(t) = \frac{\xi_f - \xi_0}{T_p}\sigma'(\lambda), \tag{90}$$

$$\ddot{\xi}_r(t) = \frac{\xi_f - \xi_0}{T_p^2}\sigma''(\lambda), \tag{91}$$

$$\dddot{\xi}_r(t) = \frac{\xi_f - \xi_0}{T_p^3}\sigma'''(\lambda). \tag{92}$$

The derivatives of the blending function are

$$\sigma'(\lambda) = 30\lambda^2 - 60\lambda^3 + 30\lambda^4, \tag{93}$$

$$\sigma''(\lambda) = 60\lambda - 180\lambda^2 + 120\lambda^3, \tag{94}$$

$$\sigma'''(\lambda) = 60 - 360\lambda + 360\lambda^2. \tag{95}$$

The polynomial satisfies

$$\sigma(0) = 0, \quad \sigma(1) = 1, \quad \sigma'(0) = \sigma'(1) = 0, \quad \sigma''(0) = \sigma''(1) = 0. \tag{96}$$

This is for a rest-to-rest phase; the scheduled variable has zero endpoint rate and zero endpoint acceleration. This property removes acceleration discontinuities at the phase boundaries and reduces the impulsive forcing that would otherwise impact the internal sloshing payload.

The use of the quintic polynomial in (14) follows directly from the unconstrained scalar minimum-jerk problem. For the cost

$$J_\xi = \int_{t_0}^{t_f} \left(\dddot{\xi}_r(t)\right)^2 dt,$$

the Euler–Lagrange condition gives

$$\frac{d^3}{dt^3}\left(\frac{\partial}{\partial \dddot{\xi}_r}\left(\dddot{\xi}_r^2\right)\right) = 0 \quad \Rightarrow \quad \xi_r^{(6)}(t) = 0. \tag{97}$$

Therefore, the minimizing trajectory is a fifth-order polynomial. Applying the endpoint position, velocity and acceleration constraints yields the closed-form blend given in Eqn. 88.

The complete reference is then generated by applying Eqn. 89 to the phase variables used in the VTOL mission. For instance, this would mean that the hover-to-cruise and cruise-to-hover transitions, the rotor tilt angle, airspeed, altitude, and heading are all blended smoothly. During the banked-turn segment in the trajectory, the heading schedule is smoothed, and the nominal roll angle is adjusted to meet the coordinated-turn relation given by

$$\phi_r = \tan^{-1}\left(\frac{V_r \dot{\psi}_r}{g}\right). \tag{98}$$

Which is subjected to the prescribed bank-angle limit, the pitch reference is then computed from the desired flight-path angle and trimmed angle of attack, and the attitude reference is assembled by

$$R_{bn,r} = R_z(\psi_r) R_y(\theta_r) R_x(\phi_r). \tag{99}$$

For the final phase of the flight regime, the landing segment fixes the lateral reference position while the vertical position is brought from the final hover altitude to the touchdown point using the same minimum-jerk blend:

$$x_{N,r}(t) = x_{N,\mathrm{hov}}, \tag{100}$$

$$y_{N,r}(t) = y_{N,\mathrm{hov}}, \tag{101}$$

$$z_{N,r}(t) = z_{N,\mathrm{hov}} + \left(z_{N,\mathrm{td}} - z_{N,\mathrm{hov}}\right)\sigma(\lambda). \tag{102}$$

Because the given $\dot{z}_r$ and $\ddot{z}_r$ are zero, at the beginning and the end of the descent, the landing command does not introduce a step change in the vertical velocity or acceleration. This then limits the variation in acceleration affecting the vehicle and, by extension, the sloshing liquid payload.

In this section, the minimum-jerk scheduling is used solely to provide a smooth kinematic phase transition; it does not guarantee that the reference is dynamically feasible for the VTOL aircraft. The schedule variables are therefore combined with lower-order aerodynamic and propulsive force-balance calculations. At each time step, the generator estimates the required lift, drag compensation, rotor tilt, and thrust, then applies the actuator and saturation limits. This separation allows the reference to be optimized to reduce jerk, while the force-balancing logic maintains compatibility with the physical model.

Using this logic, the reference for a closed-loop system is developed and presented in Fig. 3.

VIII. Discussion

A. Aircraft Model Derivation

The model developed in this work provides a control-oriented representation of a fixed-wing tilt-rotor VTOL aircraft intended for wildfire-fighting operations. The central difficulty with this type of aircraft is that it must operate across a wide flight envelope, beginning in a rotor-borne hovering operation, transitioning to a coupled regime, and ending in

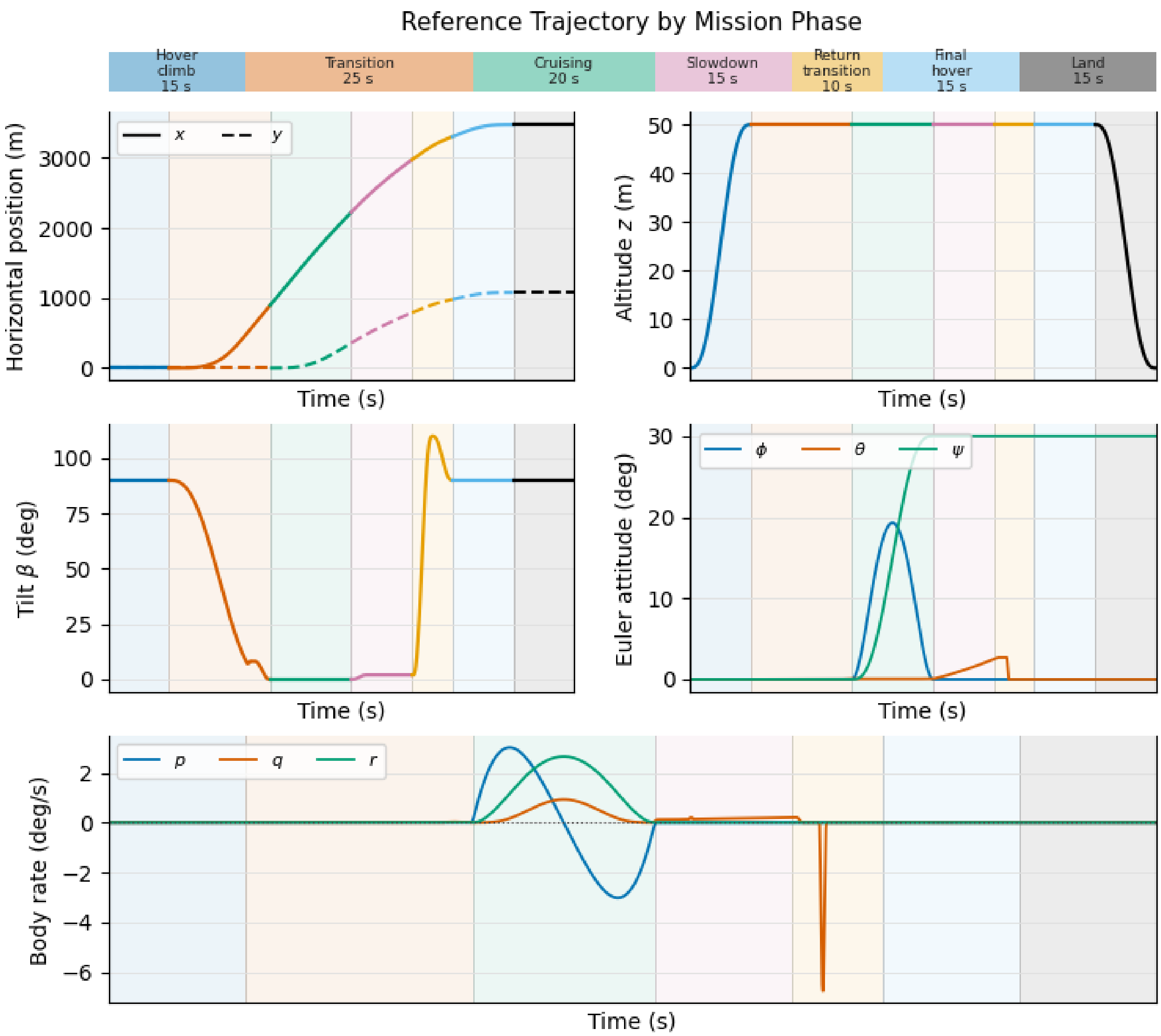

Fig. 3 VTOL Fixed-Wing Minimum Jerk Reference Generation

a wing-borne forward flight configuration. During the process, the dominant sources of lift, propulsion and control authority change continuously.

A useful transition model must maintain the main vehicle-level coupling without becoming too computationally expensive for trajectory generation or control design. The proposed formulation addresses this need by combining a six-degree-of-freedom rigid-body model with offset tilting rotors, actuator dynamics, aerodynamic force and moment models, prop wash modified control surface and lift effectiveness, and finally a low-order internal sloshing mode. Kane's method is a useful modelling tool in this setting because it provides a systematic way to include offset forces, non-conservative loads, and the moving sloshing mass in a mass matrix form. This is important for the present application because the water- or fire-retardant polyurethane is not only a mass contribution; it also introduces additional dynamic coupling when the aircraft experiences rapid changes in acceleration.

A significant feature of the model is that the rotor system and aerodynamic system are not treated as separate flight modes. Instead, rotor thrust, rotor tilt, fixed-wing lift and control-surface moments are all present in the same dynamic model. This involves a hard switch between hover and cruise modes and is more appropriate for transition flight, where the aircraft may depend simultaneously on rotor lift, wing lift, and prop-wash-affected control authority. The inclusion of thrust lag, rotor-tilt rate limits and control surface rate limits is also an important consideration because transition trajectories that appear smooth kinematically can still be infeasible if they require instantaneous changes in force or moments.

The prop was, and a local dynamic-pressure model provides a partial representation of one of the main physical effects in transition flight. In hover and low-speed flight, conventional aerodynamic surfaces may experience little freestream dynamic pressure; however, when located within the rotor slipstream, they can still generate some control authority. While this is not explicitly addressed in the paper, it should be acknowledged that there may be an impact. Thus, the use of weighted prop wash terms for the elevator, ailerons, and rudder allows this effect to be included without fully resolving the full rotor wake. These terms, as discussed earlier in the paper, are simplified; however, they still capture the important trend that rotor effectiveness and slipstream coherence vary throughout the transition envelope.

B. Reference Generation

The minimum-jerk reference generation is consistent with the needs of a sloshing-sensitive wildfire aircraft. Since abrupt acceleration changes excite the fluid payload's internal motion, the trajectory should avoid discontinuities in velocity and acceleration where possible. The quintic blending function used in this work yields zero endpoint velocity and acceleration for scheduled variables, thereby reducing artificial excitation at phase boundaries. This is useful for rotor-tilt, altitude, airspeed, heading and landing-descent schedules. This, however, is not proof of dynamic feasibility. A minimum-jerk curve may still require more thrust, lift, or control authority than the model can provide. For this reason, in the derivation, the generator combines smooth phase scheduling with lower-order force-balance calculations and actuator saturation limits to ensure the trajectory is feasible.

This resulting reference trajectory is therefore best understood as analogous to an actuator-aware, model-consistent command rather than a purely geometric path. This is useful for future control design because it provides a smooth initial reference while respecting the aircraft model's major physical limits. In particular, the reference generator reduces the likelihood that the controller will need to track abrupt changes in various states, such as altitude, speed, or rotor tilt. This is done in theory to improve the optimizer's numerical conditioning and reduce the controller's tendency to rely on saturated thrust or aggressive changes during transition.

IX. Conclusion of Model and Reference development

This paper presented a control-oriented dynamic model and reference-generation framework for a fixed-wing tilt-rotor VTOL aircraft intended for wildfire-fighting missions. The vehicle was modelled as a six-degree-of-freedom rigid body with an internal slosh mode, two tilting rotors, finite actuator bandwidth, rotor reaction torque, rotor-induced velocity, ground effect, thrust-inflow correction, and prop wash-modified aerodynamic control-surface effectiveness. The fixed-wing aerodynamic model was formulated using conventional stability and control derivatives, with additional transition-specific corrections to mitigate unrealistic aerodynamic authority outside the forward-flight regime.

A key outcome of the work is the integration of rotor-borne and wing-borne effects within a single dynamic framework. This is important for transition flight because the vehicle does not pass instantaneously from hover to cruise; instead, rotor thrust, wing lift, prop wash, and aerodynamic control moments all contribute over different portions of the maneuver. The resulting model is more detailed than a point-mass or purely kinematic transition model, while remaining simpler than blade-resolved or free-wake rotor craft simulations. This makes it suitable for trajectory generation, controller development, and future control derivation studies.

The main limitation of the presented work is that the model is reduced-order and not fully calibrated against experimental data, computational fluid dynamics results, or flight-test data. The prop wash impingement factors are treated as tunable parameters based on empirical information. The rotor-induced-flow model is based on actuator-disk momentum theory with algebraic corrections for cross-flow, thrust lapsing, and ground effect. While this is appropriate for control-oriented simulation, it cannot capture blade-resolved loading, wake roll-up, vortex-wing-state behaviour, or unsteady rotor-wing interference. Therefore, the present model should be used for controller development, trajectory feasibility studies, and sensitivity analysis rather than final aerodynamic performance prediction.

The sloshing model is also a simplified representation of the internal liquid payload dynamics. The purpose is to provide a representative approach to the forces and moments acting on the aircraft due to the sloshing of the payload. The use of a lumped mass-spring-damper model captures the first-order effect of the moving internal mass, but it does not represent the free-surface motion, tank geometric effects or nonlinear fluid impacts. For wildfire applications, the effects may become important factors during aggressive maneuvers, partial tank fill conditions or payload drop events. Future work will therefore be comparing the lumped slosh model against higher-fidelity fluid models or experimental tank data.

The main conclusion is that a smooth and physically informed reference is essential for transition-flight control of a slosh-sensitive tilt-rotor VTOL aircraft. Minimum-jerk scheduling reduces discontinuities, while the force-balance and saturation logic help ensure that the commanded transition remains compatible with the vehicle model. The framework developed here, therefore, provides a foundation for future closed-loop control implementation and for evaluating the feasibility of autonomous fixed-wing VTOL aircraft in wildfire-response missions.

X. Future Work and Development

Future work and development of this model would focus on three main improvements. Firstly, the aerodynamic and prop wash parameters should be calibrated using higher-fidelity aerodynamic tools, potential wind-tunnel data or flight-test identification. Second, the reference trajectory should be tested in a closed-loop simulation with the full-order plant and a controller, to evaluate tracking performance, actuator usage, slosh response and robustness to wind disturbances. Third, the wildfire mission profile should be expanded to include operational constraints such as gusts, terrain clearance, payload release, loitering, and repeated transition cycles. These steps would allow the present modelling framework to be evaluated not only as a dynamic model, but also as a practical basis for autonomous VTOL wildfire-response operations.

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